

A Balance of Risks

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ground truths

Select U.S. Macroeconomic Features, 2026:Q1

real GDP growth:

unemployment rate:

inflation rates:

wage growth:

interest rates:

Select U.S. Macroeconomic Features, 2026:Q1

real GDP growth: 1.6%

unemployment rate: 4.3%

inflation rates: 2.5% (Core CPI), 3.1% (Core PCE)

wage growth: 3.4% (ECI)

interest rates: 3.75% (fed funds), 4.3% (10 yr. treasury bond)

Source: Wells Fargo Economics.

inconvenient arithmetic

the time value of money

$$P_0 = \frac{R_1}{(1+i)} + \frac{R_2}{(1+i)^2} + \dots + \frac{R_n}{(1+i)^n} + \frac{P_n}{(1+i)^n}$$

where R_j is the flow of rent paid at the end of each period j ;
and i is the nominal interest rate.

the time value of money

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where R_j is the flow of rent paid at the end of each period j ;
and i is the nominal interest rate.

$$i \downarrow \iff P_0 \uparrow$$

$$i \uparrow \iff P_0 \downarrow$$

credit channel

Figure 1: U.S. Treasury 10-Year Yield to Maturity



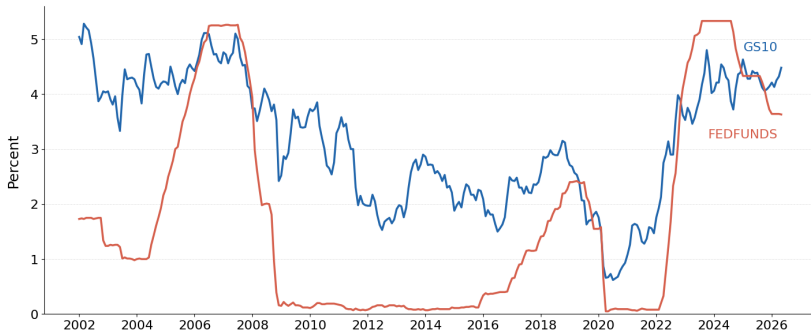
Source: Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis.

Figure 2: U.S. Treasury 10-Year Yield to Maturity



Source: Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis.

Figure 3: U.S. Treasury 10-Year Yield to Maturity



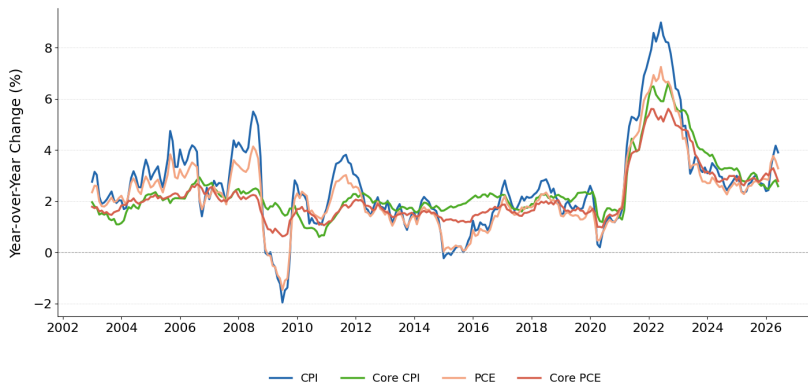
Source: Federal Reserve Economic Data (FRED), Federal Reserve Bank of St. Louis.

independent thought



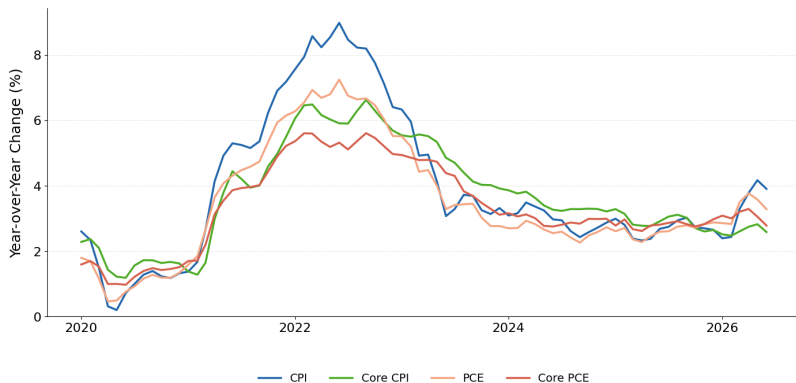
Kevin Warsh

Figure 4: U.S. Inflation



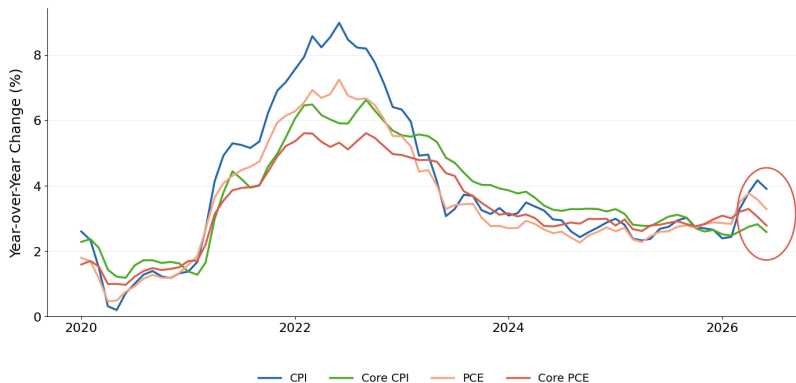
Source: Bureau of Labor Statistics; Bureau of Economic Analysis, via FRED.

Figure 5: U.S. Inflation



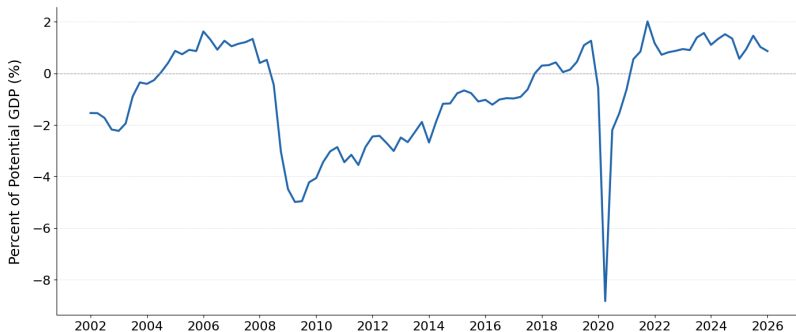
Source: Bureau of Labor Statistics; Bureau of Economic Analysis, via FRED.

Figure 6: U.S. Inflation



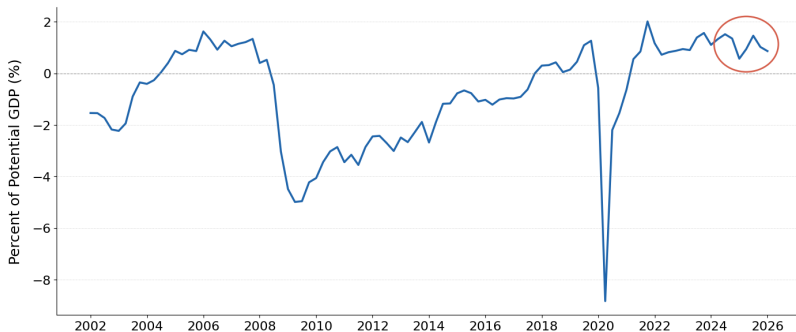
Source: Bureau of Labor Statistics; Bureau of Economic Analysis, via FRED.

Figure 7: U.S. Output Gap



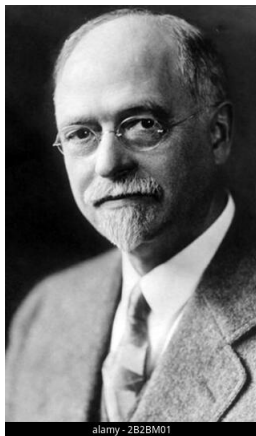
Source: Congressional Budget Office; Bureau of Economic Analysis, via FRED. Output gap = $(GDPC1 - GDPPOT) / GDPPOT$.

Figure 8: U.S. Output Gap



Source: Congressional Budget Office; Bureau of Economic Analysis, via FRED. Output gap = $(GDPC1 - GDPPOT) / GDPPOT$.

go fish



alamy - 2B2BM01

Irving Fisher (1867 - 1947)

the Fisher effect

$$i = r + \pi^e$$

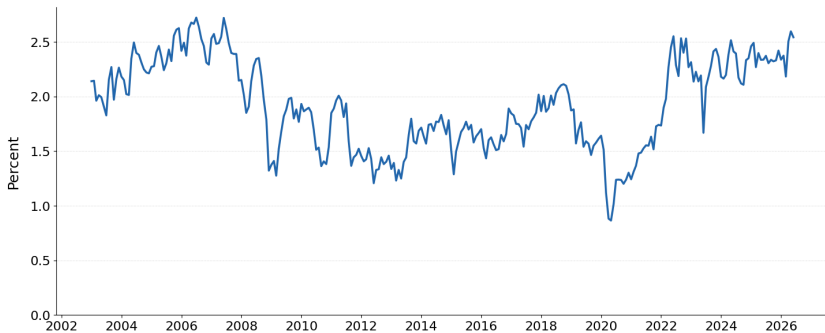
where i is the nominal interest rate,
 r is the real interest rate,
 π^e is the expected inflation rate,
and $*$ denotes an equilibrium value.

the Fisher effect

$$i = r + \pi^e$$

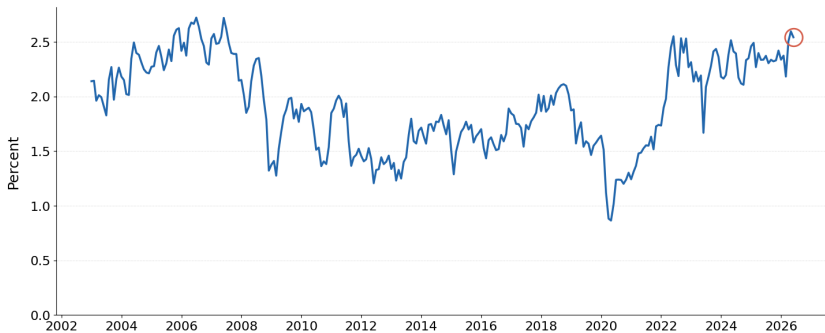
where i is the nominal interest rate,
 r is the real interest rate,
 π^e is the expected inflation rate,
and $*$ denotes an equilibrium value.

Figure 9: U.S. 5-Year Expected Inflation



Source: Federal Reserve Bank of Cleveland, via FRED. Series: EXPINF5YR.

Figure 10: U.S. 5-Year Expected Inflation



Source: Federal Reserve Bank of Cleveland, via FRED. Series: EXPINF5YR.

the Fisher effect

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the Fisher effect

$$i = r + \pi^*$$

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$$i = r + \pi^*$$

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the Fisher effect

$$i^* = r^* + \pi^*$$

where i is the nominal interest rate,
 r is the real interest rate,
 π^e is the expected inflation rate,
and $*$ denotes an equilibrium value.

r-star

$$r^* = f(\rho, g, \delta, s)$$

where r^* is the natural rate of interest,
 ρ is the coefficient of time preference,
 g is the rate of productivity growth,
 δ is the age of the population,
and s is the global saving rate.

r-star

$$r^* = f\left({}^+\rho, g, \delta, s\right)$$

where r^* is the natural rate of interest,
 ρ is the coefficient of time preference,
 g is the rate of productivity growth,
 δ is the age of the population,
and s is the global saving rate.

r-star

$$r^* = f\left(\rho, g^+, \delta, s\right)$$

where r^* is the natural rate of interest,
 ρ is the coefficient of time preference,
 g^+ is the rate of productivity growth,
 δ is the age of the population,
and s is the global saving rate.

r-star

$$r^* = f\left(\rho, g, \bar{\delta}, s\right)$$

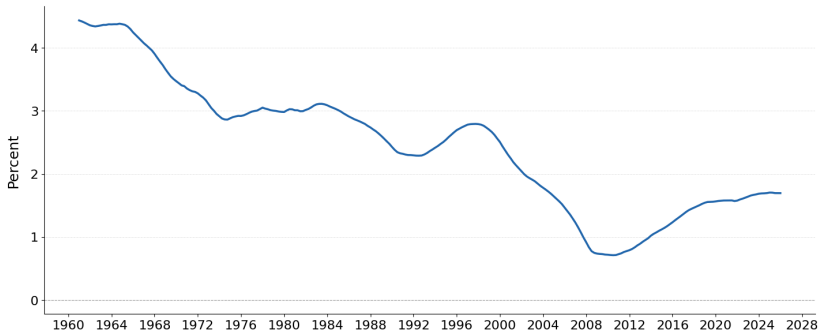
where r^* is the natural rate of interest,
 ρ is the coefficient of time preference,
 g is the rate of productivity growth,
 $\bar{\delta}$ is the age of the population,
and s is the global saving rate.

r-star

$$r^* = f\left(\rho, g, \delta, \bar{s}\right)$$

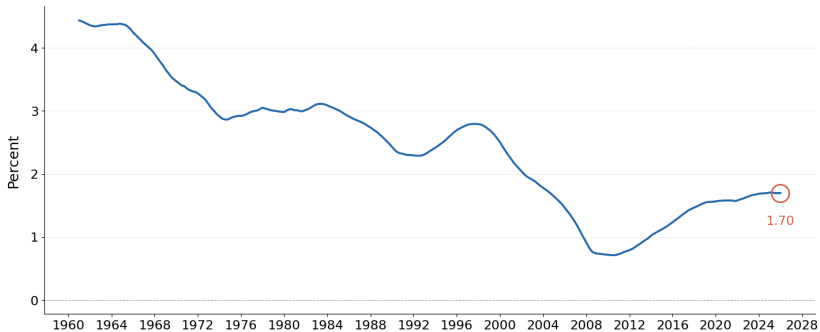
where r^* is the natural rate of interest,
 ρ is the coefficient of time preference,
 g is the rate of productivity growth,
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and s is the global saving rate.

Figure 11: U.S. Estimate of r^*



Source: Laubach and Williams, Federal Reserve Bank of New York. Two-sided estimate.

Figure 12: U.S. Estimate of r^*



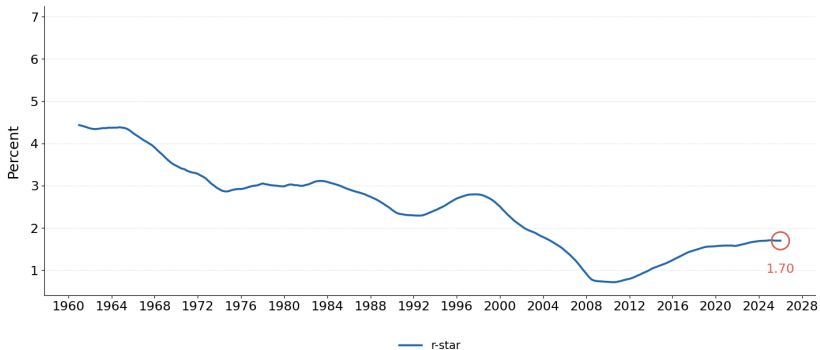
Source: Laubach and Williams, Federal Reserve Bank of New York. Two-sided estimate.

the Fisher effect

$$i^* = r^* + \pi^*$$

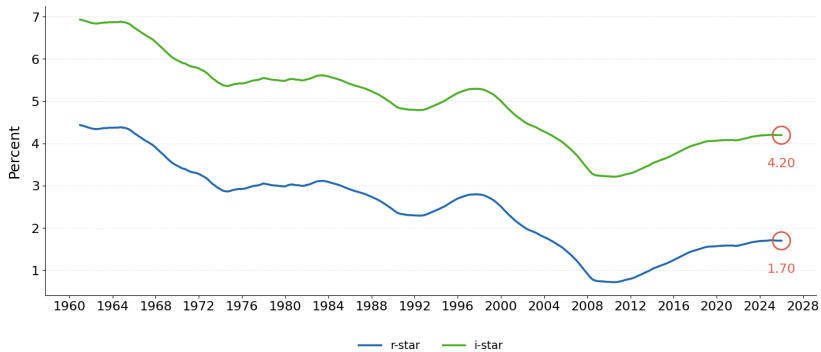
where i is the nominal interest rate,
 r is the real interest rate,
 π^e is the expected inflation rate,
and $*$ denotes an equilibrium value.

Figure 13: U.S. Estimate of r^*



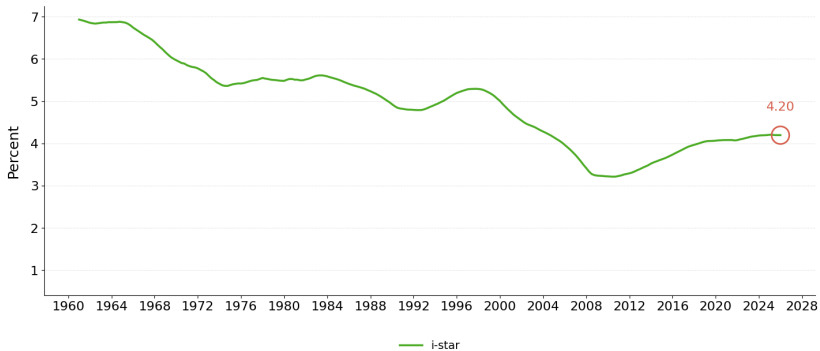
Source: Laubach and Williams, Federal Reserve Bank of New York. Two-sided estimate.

Figure 14: U.S. Estimate of r^* and i^*



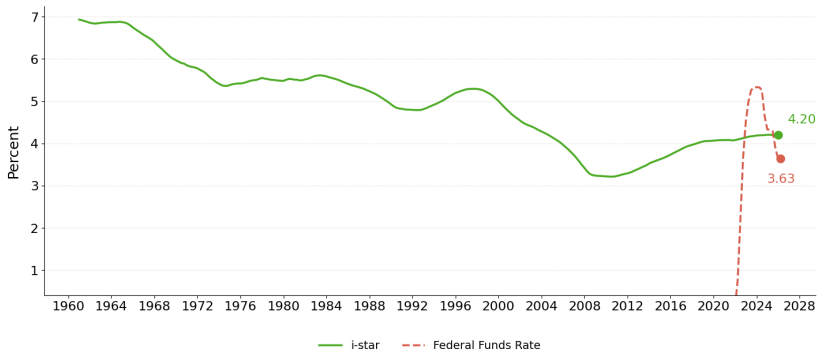
Source: Laubach and Williams, Federal Reserve Bank of New York. Two-sided estimate.

Figure 15: U.S. Estimate of r^* and i^*



Source: Laubach and Williams, Federal Reserve Bank of New York. Two-sided estimate.

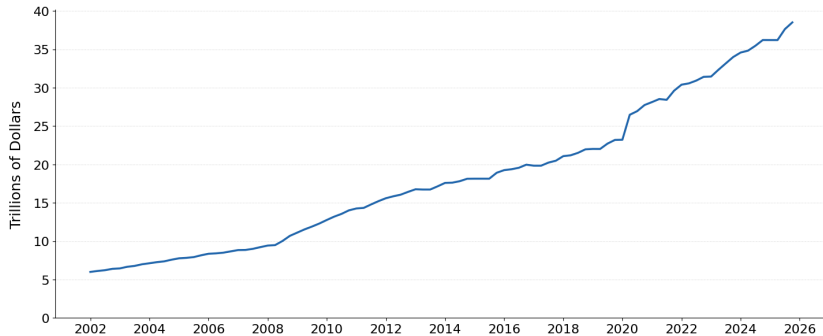
Figure 16: U.S. Estimate of r^* and i^*



Source: Laubach-Williams (FRBNY); Federal Reserve (FRED). Two-sided estimate.

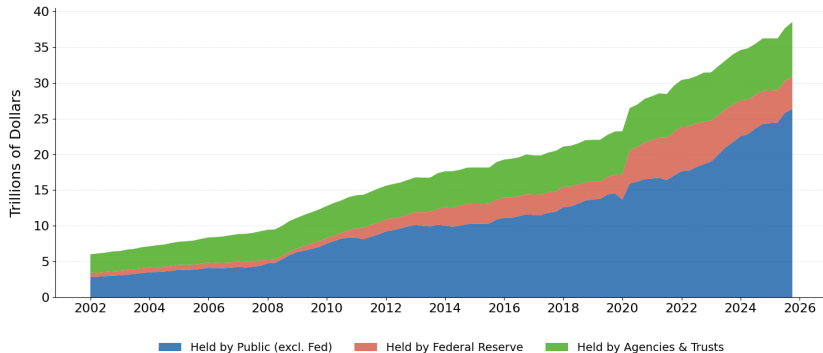
buried treasury

Figure 17: U.S. Federal Debt Outstanding



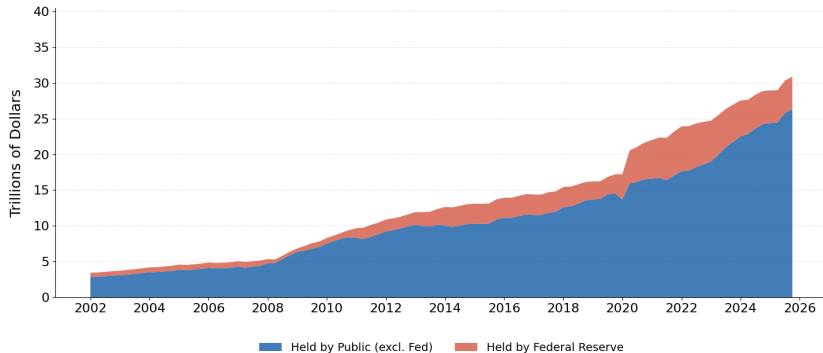
Source: U.S. Treasury, via FRED. Series: GFDEBTN.

Figure 18: U.S. Federal Debt Outstanding



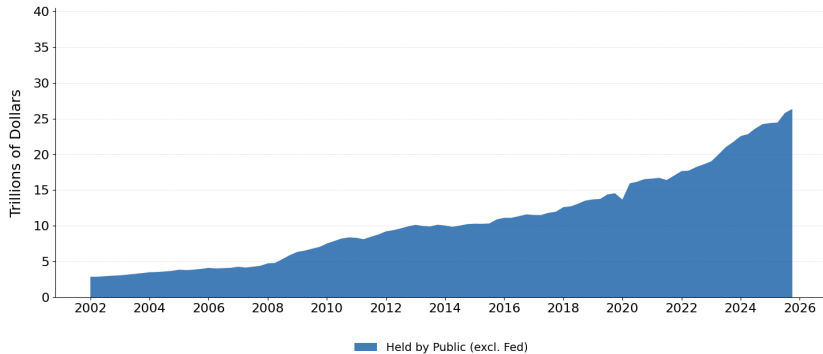
Source: U.S. Treasury; Federal Reserve, via FRED. GFDEBTN, FYGFDPUN, FDHBFBRBN, FDHBATN.

Figure 19: U.S. Federal Debt Outstanding



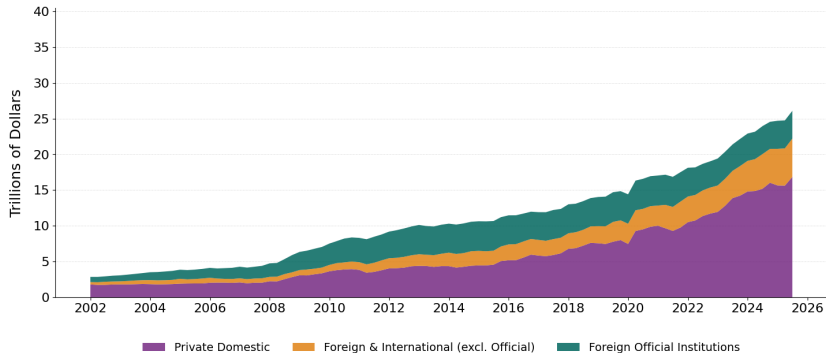
Source: U.S. Treasury; Federal Reserve, via FRED. FYGFDPUN, FDHBFBRBN.

Figure 20: U.S. Federal Debt Outstanding



Source: U.S. Treasury; Federal Reserve, via FRED. FYGFDUN, FDHBFBN.

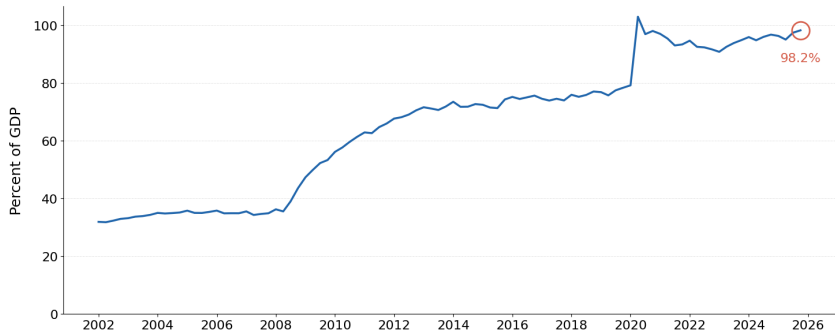
Figure 21: U.S. Federal Debt Outstanding



Source: U.S. Treasury; Federal Reserve; BIS, via FRED. FDHBPIN, FDHBFIN, BOGZ1FL263061130Q.

an inconvenient truth

Figure 22: U.S. Federal Debt Held by the Public



Source: U.S. Treasury; Bureau of Economic Analysis, via FRED. Series: FYFGDQ1885.

debt dynamics

$$debt \uparrow \Leftrightarrow deficit \uparrow$$

$$debt \uparrow \Leftrightarrow interest\ rates \uparrow$$

$$debt \uparrow \Leftrightarrow GDP \downarrow$$

Where *debt* and *deficit* are relative to GDP.

debt dynamics

debt ↑ ⇔ *deficit* ↑

debt ↑ ⇔ *interest rates* ↑

debt ↑ ⇔ *GDP* ↓

Where *debt* and *deficit* are relative to GDP.

debt dynamics

$$debt \uparrow \Leftrightarrow deficit \uparrow$$

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$$debt \uparrow \Leftrightarrow GDP \downarrow$$

Where *debt* and *deficit* are relative to GDP.

debt dynamics

debt ↑ ⇔ *deficit* ↑

debt ↑ ⇔ *interest rates* ↑

debt ↑ ⇔ *GDP* ↓

Where *debt* and *deficit* are relative to GDP.

Takeaways

inconvenient arithmetic: $i \uparrow \Rightarrow p \downarrow$

r-star is born: $r^* \uparrow \Rightarrow i \uparrow$

buried treasury: $\frac{\text{debt}}{\text{gdp}} \uparrow \Rightarrow i \uparrow$

Taylor rules: $i < i_{\text{target}} \Rightarrow i \uparrow$

state headwinds: $\frac{\overset{\uparrow}{\text{income}}}{\text{pop}} = \frac{\text{income}}{\text{employed}} \times \frac{\overset{\uparrow}{\text{employed}}}{\text{labor force}} \times \frac{\overset{\uparrow}{\text{labor force}}}{\text{pop}}$

thank you

...and save the date:

October 15, 2026 at the Sanford PREMIER Center, Sioux Falls

[SDLLC.2026](#)

